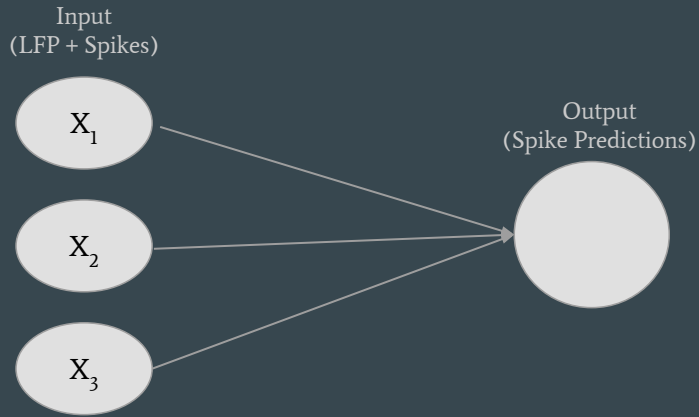


Predicting Neuron Spiking from LFP Data : The Role of Model Complexity

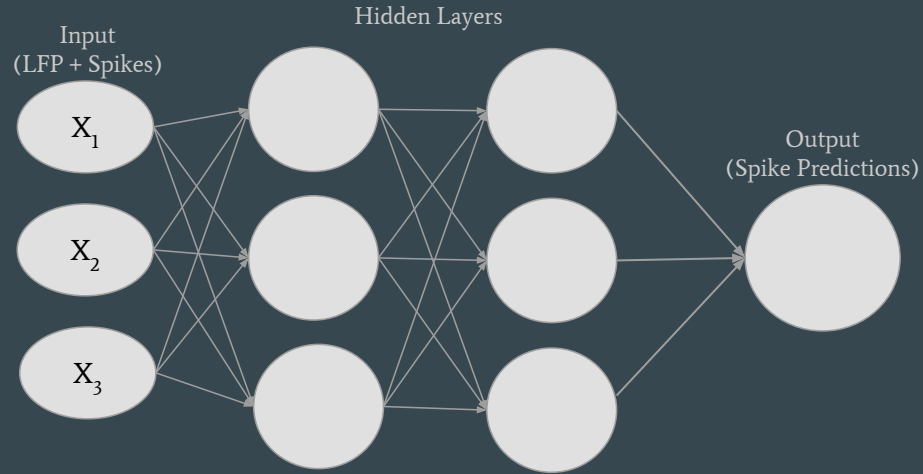


Kai, Eitan, Winnie
12/11/2025

Why Model Complexity?



Simple Model : GLM - Linear, Fewer Parameters



Complex Model : Neural Network - Non-Linear, Many Parameters

Background: Modern Models VS GLMs for Spikes Prediction

- Spikes carry information about behavior and cognition
- LFPs are slow summed synaptic signals that reflect population activity and contain oscillations (theta, gamma) linked to spike timing (Banerjee et al. 2012; Hall et al. 2014; Zhou et al. 2015).
- Recent benchmarks show modern ML models (neural nets, boosted trees) often predict spikes more accurately than Poisson GLMs (Benjamin et al. 2018).

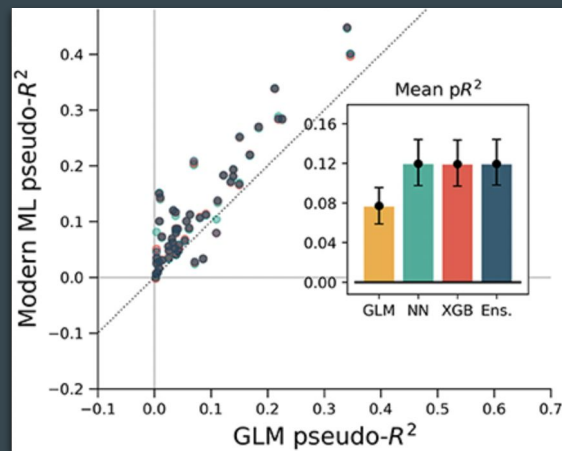


Figure from Benjamin et al., 2018 ("Modern Machine Learning as a Benchmark for Fitting Neural Responses")

Does model complexity improve the prediction of a neuron's spiking based on LFP data?

Specifically: Does a neural network predict spikes better than a Poisson GLM?

Hypothesis

Given the same LFP + spike-history features, a neural network predicts spikes better than a Poisson GLM, indicating that additional model complexity captures what the simpler model misses.

Methods : Data Processing

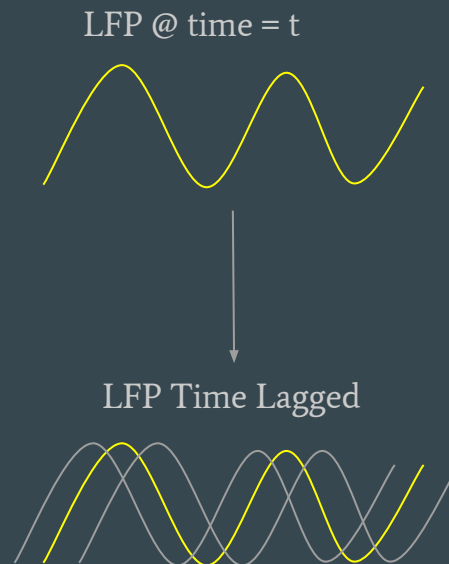
Dataset : Running Data from Hippocampal LFP and spike trains from a rat in a three-arm maze task.

Sampling : 25 Hz

Inputs (X) : Toeplitz design matrix from LFP

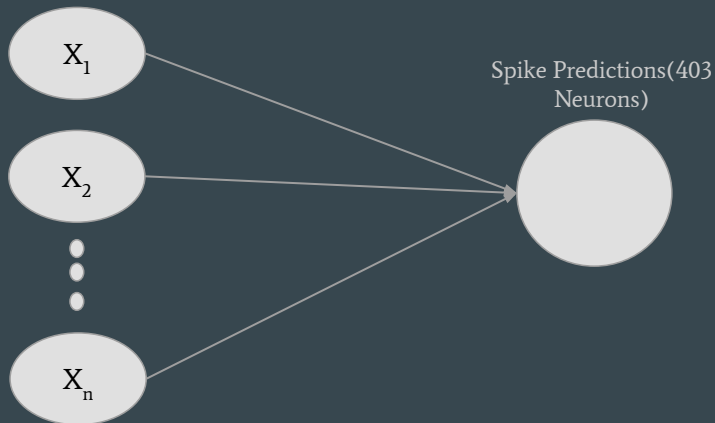
Outputs (Y) : Spike counts for each neuron

PreProcessing : Split data into train and test sets to prevent overfitting. Standardize and scaled to make resistant to outliers.



Our Model's Architecture

Input
(960 LFP Features)



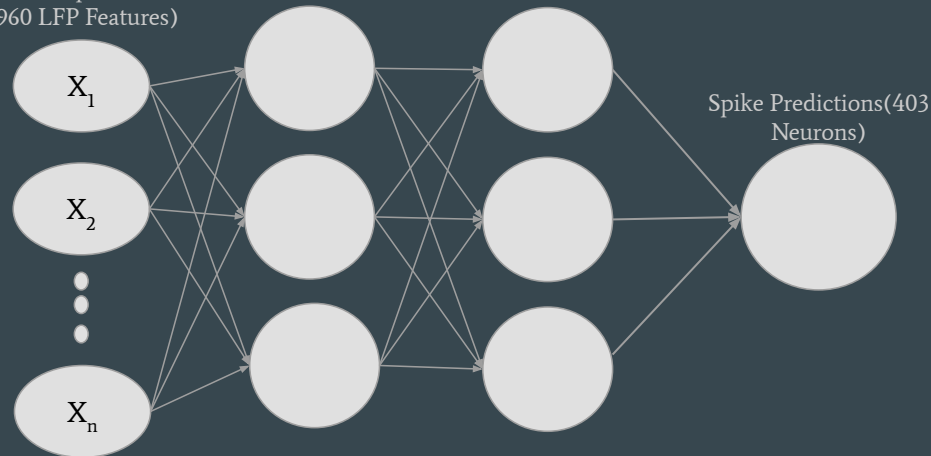
- One linear step from 960 LFP features to 403 neurons
- Exponential link to keep spike rates positive
- Trained with Poisson loss on the same train/test split as the NN

2 Hidden Layers

Input
(960 LFP Features)

128 Nodes

512 Nodes



- $960 \rightarrow 128 \rightarrow 512 \rightarrow 403$ units with ReLU in the hidden layers
- Dropout on hidden layers for regularization
- Same LFP inputs and Poisson loss as the GLM, but with multiple processing stages

Neural Network Parameter Decisions

Loss: Poisson NLL - Matches discrete spike counts.

Optimizer: AdamW - Adam and weight decay for more stable training. Loshchilov, I., & Hutter, F. (2019). Decoupled Weight Decay Regularization. *International Conference on Learning Representations (ICLR)*.

Hidden nonlinearity: ReLU is simple, effective for nonlinear spike/LFP mapping.

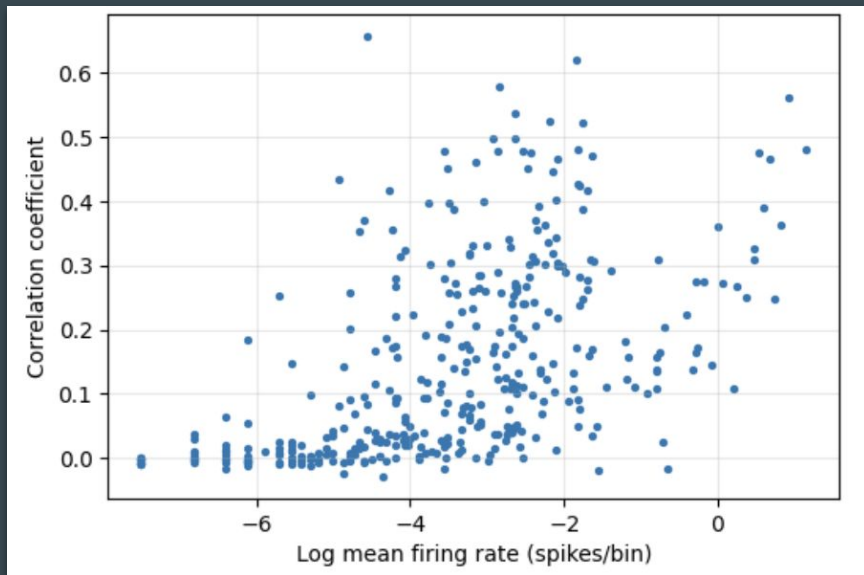
Output: Softplus - Smooth, strictly positive firing-rate predictions.

Training + Hyperparameters

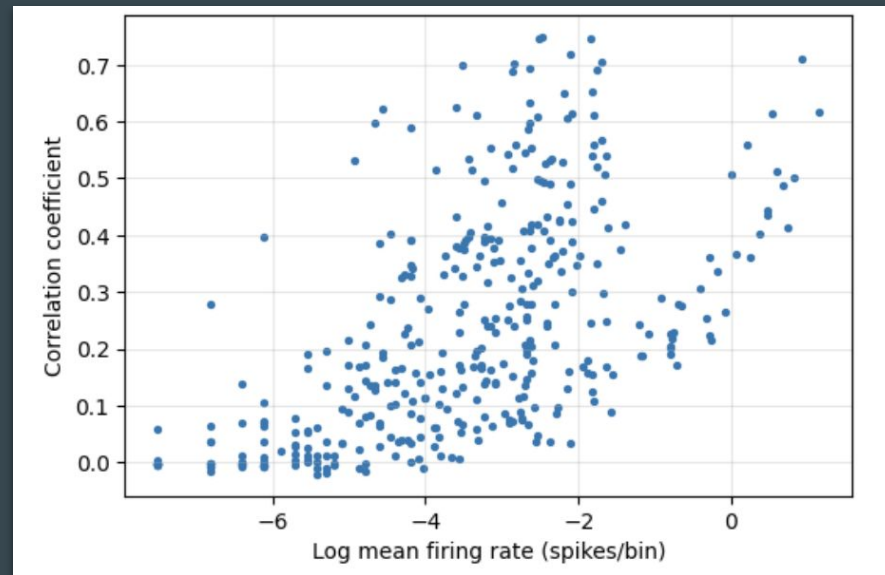
- Hyperparameter Search method: Used the Python library Optuna to automatically tune learning rate, dropout, and layer sizes
- Best NN configuration (Optuna):
 - Learning rate: 0.000556
 - Layer 1: 128 units, dropout 0.106
 - Layer 2: 512 units, dropout 0.496
 - Hidden layers: 2

Results - Performance vs Firing Rate

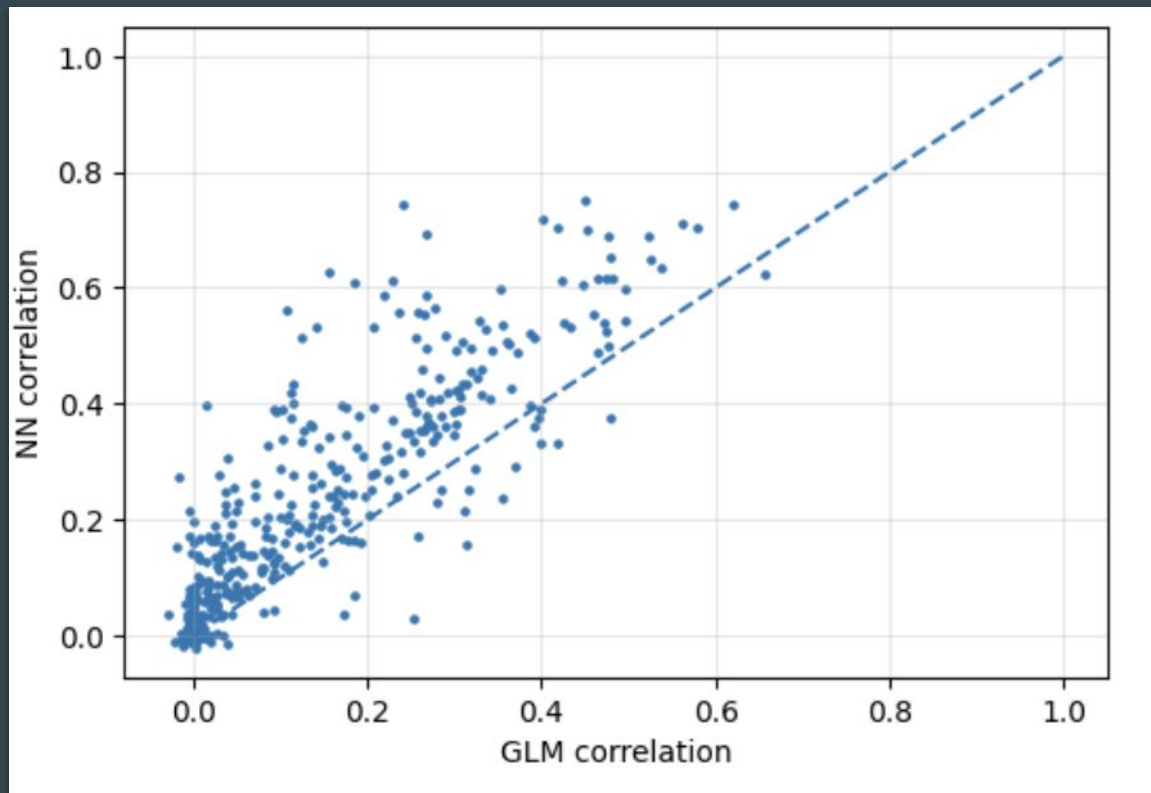
GLM Correlation VS Mean Firing Rate



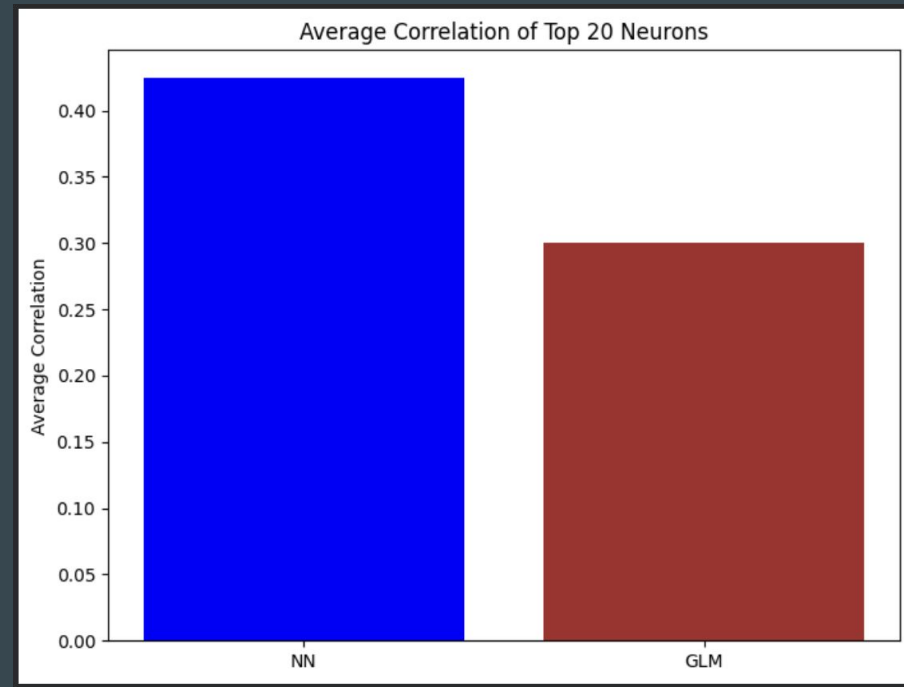
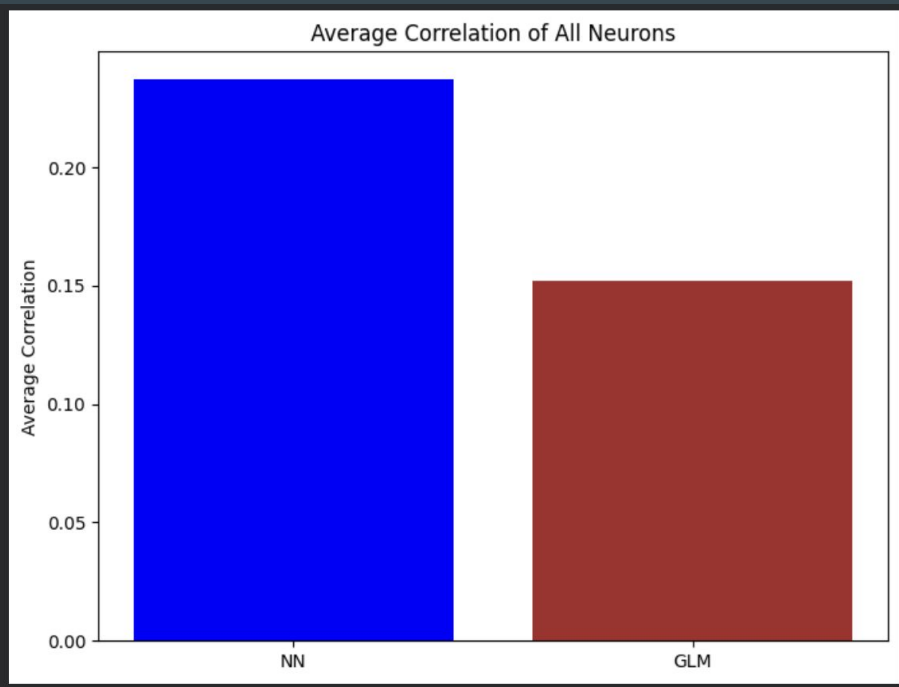
NN Correlation Vs Mean Firing Rate



Results: Neural Network vs GLM (Per Neuron)



Results: Neural Network vs GLM



Discussion

Hypothesis : A neural network should predict spikes more accurately than a Poisson GLM using the same LFP features

For most neurons, NN test correlations are higher than GLM correlations (points lie above the diagonal in our scatter plot)

Nonlinear hidden layers captures complex relationships between LFP and spikes, whereas the GLM is limited to a single linear transformation

Biology Connection: Brain Complexity Relation to Layers

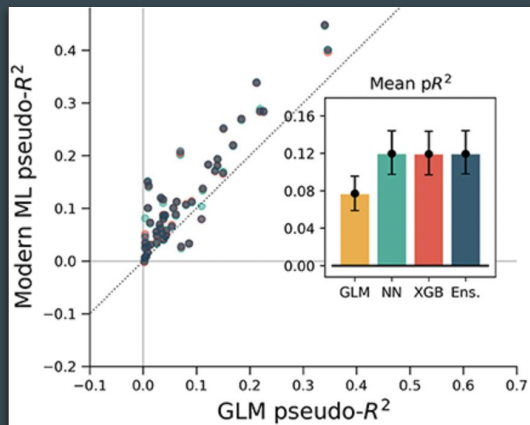
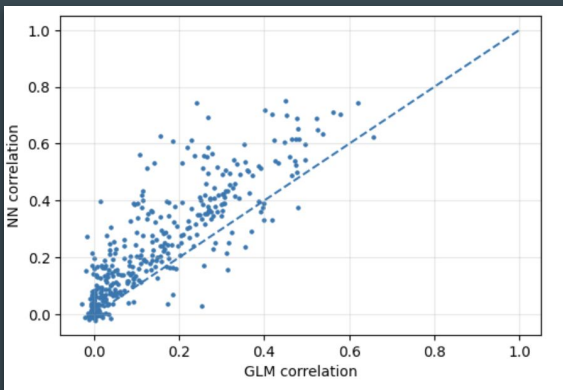


Figure from Benjamin et al., 2018 ("Modern Machine Learning as a Benchmark for Fitting Neural Responses")

Limitations and Further Study

Apply the same GLM vs NN comparison to other brain regions to assess generalizability

Add spike-history and behavioral features (maze position) to see if NN still beats GLM

Explore larger / alternative architectures and other ML models (boosted trees, RNNs)

QUESTIONS?

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